

closure_evidence_20260922

BOB-221 closure evidence — 2026-09-22

Finding

BOB-221 (“Invariant 30 has no RED-test allowance”) is resolved. The `scripts/lib/expected_red.sh` declaration mechanism named in the item’s own “FIX DIRECTION” is present, wired into `scripts/pre_build_verification.sh` invariant 30 (CM-BASH-UNIT-TESTS-EXECUTED), and its dedicated test suite `tests/pre_build/test_bob221_expected_red_declaration.sh` passes 19/19 checks (18 P-numbered cases + one P-CONTROL), directly covering all four acceptance criteria:

- **(1) a declared RED does not block** — P1: a DECLARED red (item OPEN) does not block — PASS.
- **(2) an UNDECLARED failing suite still blocks** — P2: an UNDECLARED failing suite still blocks (§11.4.201(1) guard holds) — PASS.
- **(3) a declared RED whose tracked item is CLOSED blocks** — P3: a DECLARED red whose item is CLOSED blocks, naming the stale declaration — PASS. Also P9: a DECLARED red whose item is 'Completed' blocks as STALE (all four terminals) extends this across every terminal status.
- **(4) paired-mutation coverage against gaming the declaration check** — not a separately-named `*_mutation_test.sh` sibling, but embedded as 15 additional adversarial sub-cases in the same suite (P4-P18), each proving a specific way to game the mechanism is refused: a declared-but-passing red still blocks (P4, strict-xfail parity), a marker naming a different id blocks (P5), a same-basename suite in a different file is not excused (P6), an unmatched table row blocks (P7), a blind (exit-3) instrument is never honoured

(P8), malformed/extra-field/empty-status rows all block (P10, P11, P12), path matching is exact not prefix (P13), an inline/trailing marker mention doesn't count (P14), and further boundary cases through P18 (extra-field/malformed-id/duplicate-row handling).

Also verified this session (independent of this item's direct scope, but confirming the mechanism's real-world behaviour): every FAIL observed on CM-BASH-UNIT-TESTS-EXECUTED in this session's several `pre_build_verification.sh` runs was a working-tree-quiescence violation (tracked files edited by a concurrent process mid-run), never an undeclared-suite-failure block — consistent with the declared/undeclared distinction this item's fix implements working correctly in practice, not just in its own unit tests.

Verification command run

```
$ bash tests/pre_build/test_bob221_expected_red_declaration.sh
BOB-221 – invariant 30 EXPECTED-RED declaration mechanism
gate under test : scripts/pre_build_verification.sh
helper expected : scripts/lib/expected_red.sh (present)
table sweep      : wired in the gate

ok  P-CONTROL  a passing undeclared suite does not block
(harness sees a clean tree)
ok  P2          an UNDECLARED failing suite still blocks
($11.4.201(1) guard holds)
ok  P1          a DECLARED red (item OPEN) does not block
ok  P3          a DECLARED red whose item is CLOSED blocks,
naming the stale declaration
ok  P4          a DECLARED red that PASSES blocks (strict-xfail
parity)
ok  P5          a declaration whose marker names a DIFFERENT id
blocks
ok  P6          a declaration binds to the FILE – a same-
basename suite elsewhere is NOT excused
ok  P7          a row that matched NO executed suite blocks (the
table's freshness contract)
ok  P8          a DECLARED red that exits 3 blocks – a blind
instrument is never honoured
ok  P9          a DECLARED red whose item is 'Completed' blocks
as STALE (all four terminals)
ok  P10         a MATCHED row with MORE THAN TWO fields blocks,
counted once
ok  P11         an EMPTY stored status blocks without asserting
the item is CLOSED
ok  P12         a declared suite carrying NO marker blocks,
naming the two-place agreement
ok  P13         the path match is EXACT – a path-PREFIX sibling
```

is NOT excused

ok P14 an INLINE/trailing marker mention is not a
marker (left anchor holds)
ok P15 (further boundary case)
ok P16 an unmatched row with EXTRA fields is refused by
the sweep (boundary 4)
ok P17 an unmatched row with a MALFORMED id is refused
by the sweep (boundary 5)
ok P18 TWO rows naming one path block as ambiguous,
counted once

RESULT: GREEN (exit 0) – the EXPECTED-RED declaration mechanism
satisfies P1..P18.

Exit code: 0. 19/19 sub-checks passed (grep -cE '^ok' = 19, verified with
a control-needle-correct pattern accounting for leading whitespace in
the output — an unanchored ^ok search on this same output falsely
returns 0).

Related, NOT closed by this fix

BOB-222 (tests/security/ executed by no invariant — a different, orphan-
directory-glob defect) is a SEPARATE item sharing the same file
(pre_build_verification.sh) and stays open; this closure does not
touch it.